

CT scan reconstruction using machine learning

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abstract

This will be the abstract

Introduction

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Theory

Basics CT scans

The basic principle of a CT scan, is to send x-rays through a body and record the spectrum on the other side. The spectrum is shaped by how much of the x-rays are being absorbed or scattered by the body. The linear attenuation coefficient μ states how much radiation is absorbed and/or scattered per unit thickness. The intensity I recorded at the detector is given by

$$I = I_0 \cdot \exp\left(-\int \mu(x, E) dx\right) \quad (1)$$

where I_0 is the intensity of the x-rays upon entering the body and the integral in the exponent represents the cumulative attenuation $\mu(x, E)$ of the x-ray beam along its path x , where we assume $\mu(x, E)$ can change throughout the body.

Something about μ changing throughout the body To be able to simulate a body in a CT scan, material decomposition (Mechlem et al., 2018) can be used. The linear attenuation coefficient at location x will be

$$\mu(E) = \sum_{b=1}^B A_b f_b(E) \quad (2)$$

where A_b is the line integral of material b and $f_b(E)$ describes the attenuation coefficient of the basis material b . Combining the equation (??) and (2) gives an expression for the photon count at the detector:

something relating I to y_i

$$\hat{y}_i = \int_0^\infty \phi_{\text{eff},i}(E) \exp\left(-\sum_{b=1}^B A_b^i f_b(E)\right) dE \quad (3)$$

During a CT-scan $\hat{y}_i$ is measured for a set of angles. Each measurement results in a 1D array for a 2D slice.

The amount of photons detected can be plotted for each rotation angle as seen in Figure 1, where the darker areas represent less photons detected at a y -level of the detector for a specific angle.

Recent breakthroughs in photon counting detectors make use of equation (2), where the linear attenuation coefficient depends on the energy level. The photon counting detectors (pcd) allow the classification of measured photons into energy bins (Taguchi et al., 2022).

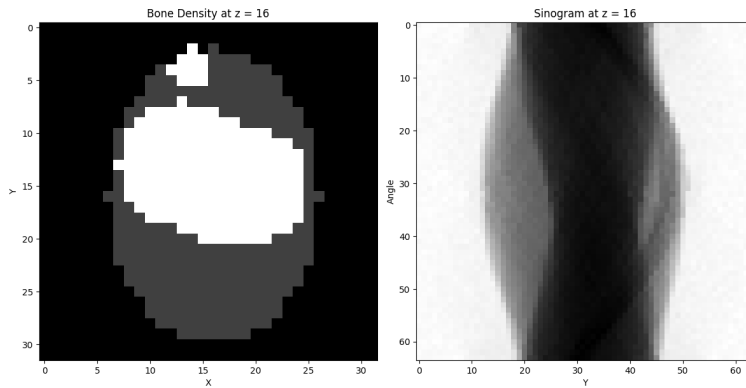


Figure 1. *left:* a 2D slice of the phantom the colour gradient representing bone density. *right* corresponding sinogram with 64 projection angles and a detector size in the y-direction of 64 pixels

Projection algorithms

- brief overview backprojection (Zeng, 2001)
- list issues
- brief overview iterative approach (with references to Mechlem)
- positives
- negatives

part of abstract: "g. They require a specific reconstruction process, consisting of two steps: material decomposition and tomographic reconstruction. Image-based methods perform reconstruction, then decomposition, while projection-based methods perform decomposition first, and then reconstruction. As an alternative, 'one-step inversion' methods have been proposed, which perform decomposition and reconstruction simultaneously. Unfortunately, one-step methods are typically slower than their two-step counterparts, and in most CT applications, reconstruction time is critical. This paper therefore proposes to compare the convergence speeds of five one-step algorithms." (Mory et al., 2018)

iterative unet (Ge et al., 2023)

machine learning - CNN

- How in image processing CNN have been used before (Yamashita et al., 2018)
- UNet github(Ronneberger, Fischer, and Brox, 2015)

Machine learning - Attention

- How we want to map the sinogram to the image, which is non-local (Vaswani et al., 2017)

Method

Data collection

- pre existing algorithm
- iterative steps
- sinogram

Pytorch

-

results

plot about speedup (time vs pixel size)

either 2D or 3D

plot about results for 32x32 simple phantom

plot harder phantom (3 features or more)

Discussion

non-machine learning speedup

pytorch is highly optimized, mechlem is not

pixel size and number of projections necessary

There will not be a linear relationship between the amount of projections necessary and the amount of pixels in the image. will this have an impact?

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